

An Additive Ansatz for the Coupled DL–AL Helmholtz Problem: $F(r, \theta) = A(r) + B(r) \Theta(\theta)$ in Place of the Product $R(r)\Theta(\theta)$

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Abstract

The problem statement (`2d-problem.tex`) and the audit note (`analytical.attempt.coupled.DL.AL.tex`) both attack the in-wedge Helmholtz eigenproblem with the multiplicative separation $F(r, \theta) = R(r)\Theta(\theta)$ and find it inconsistent with the microtubule (MT) boundary condition. Here we test a different and physically motivated trial: the *additive* ansatz

$$F(r, \theta) = A(r) + B(r) \Theta(\theta),$$

suggested by the numerical observation that the density is well described by $g(r) + a(r) \cos N\theta$ — a dominant angularly symmetric part plus a small N -fold modulation. We substitute it into the Helmholtz equation and show that the substitution *separates cleanly*: it forces $\Theta'' + \nu^2 \Theta = 0$ and sends A and B to Bessel equations of orders 0 and ν respectively, both at the *same* σ . The crucial consequence is structural: the additive ansatz is not more general than separation of variables — it is exactly the two-term Fourier–Bessel partial sum (the $\nu = 0$ mode plus the ν mode), and its only genuinely new ingredient relative to a single product is the angular-mean channel $A(r) = J_0(\sqrt{\sigma}r)$, i.e. the numerically dominant $g(r)$. We then apply the boundary conditions and find that the ansatz fails to deliver an exact single- σ eigenmode, but it fails *transparently*: the outer absorbing condition alone forces $\sqrt{\sigma}$ to be a common positive zero of J_0 and J_ν , which Bourget’s (proven) hypothesis forbids. The two channels demand mutually incompatible decay rates. This localizes the obstruction at the *outer* boundary — one line, before the MT condition is even invoked — and shows the additive ansatz to be precisely the analytic skeleton of the two-mode (Laplace/secular) reduction that does work.

1 Setup and the new ansatz

We work with the reduced wedge eigenproblem exactly as stated in `2d-problem.tex`, Eqs. (1)–(4) below. With $(\phi, \rho) = (F(r, \theta), R(r))e^{-\sigma t}$, the diffusive layer (DL) obeys the Helmholtz equation

$$-\sigma F = \nabla^2 F = F_{rr} + \frac{1}{r} F_r + \frac{1}{r^2} F_{\theta\theta}, \quad r \in [0, 1], \quad \theta \in [0, \frac{2\pi}{N}], \quad (1)$$

with the advective layer (AL) eliminated as the linear functional

$$R(r) = R[F(\cdot, 0)](r) = -\frac{a}{v} e^{\frac{b-\sigma}{v}r} \int_1^r e^{-\frac{b-\sigma}{v}r'} F(r', 0) dr', \quad (2)$$

and boundary conditions

$$(i) \ F(1, \theta) = 0, \quad (ii) \ F(r, 0) = F(r, \frac{2\pi}{N}), \quad (3)$$

$$(iii) \ a F(r, 0) = \frac{2}{r} \partial_\theta F|_{\theta=0} + b R(r), \quad (iv) \ R(1) = 0, \quad (4)$$

together with regularity at $r = 0$. (Condition (iii) uses the reflection symmetry $-\partial_\theta F|_{2\pi/N} = \partial_\theta F|_{0\cdot}$)

Rather than the multiplicative trial $F = R(r)\Theta(\theta)$, we substitute the **additive ansatz**

$$\boxed{F(r, \theta) = A(r) + B(r) \Theta(\theta)} \quad (5)$$

into (1). The motivation is empirical: the computed DL density is captured to four-digit energy accuracy by $g(r) + a(r) \cos N\theta$, a *sum* of an angularly symmetric profile and a single N -fold modulation, not by any single product. The ansatz (5) is the natural analytic representative of that form, with $A \leftrightarrow g$ and $B\Theta \leftrightarrow a \cos N\theta$.

2 Substitution into the Helmholtz equation

The Laplacian acts linearly on the two pieces of (5). Since A is angle-independent,

$$\nabla^2 A = A'' + \frac{1}{r} A', \quad \nabla^2(B\Theta) = \left(B'' + \frac{1}{r} B'\right)\Theta + \frac{1}{r^2} B \Theta''. \quad (6)$$

Inserting into $-\sigma F = \nabla^2 F$ and collecting terms,

$$\underbrace{\left\{A'' + \frac{1}{r} A' + \sigma A\right\}}_{\text{angle-independent}} + \underbrace{\left\{B'' + \frac{1}{r} B' + \sigma B\right\}\Theta + \frac{1}{r^2} B \Theta''}_{\text{angle-dependent}} = 0. \quad (7)$$

The angular factor is forced. The bracketed angle-dependent part contains $\Theta(\theta)$ and $\Theta''(\theta)$. For (7) to hold at every θ with r free, isolate the θ -dependence: dividing the angle-dependent group by $B(r)$ and demanding it be a function of θ that cancels against an r -only quantity gives the usual Sturm-Liouville separation,

$$\frac{\Theta''(\theta)}{\Theta(\theta)} = -\nu^2 = \text{const}, \quad \implies \quad \Theta'' + \nu^2 \Theta = 0. \quad (8)$$

Equivalently: 1 and $\Theta(\theta)$ are linearly independent functions of θ (as long as Θ is non-constant), so a relation of the form $c_1(r) \cdot 1 + c_2(r) \Theta(\theta) + c_3(r) \Theta''(\theta) = 0$ holding for all θ forces $\Theta'' \propto \Theta$, i.e. (8), and then forces each r -coefficient to vanish separately. This yields three decoupled ordinary differential equations.

(a) Angular equation.

$$\Theta'' + \nu^2 \Theta = 0 \quad \implies \quad \Theta(\theta) = \alpha \cos \nu \theta + \beta \sin \nu \theta. \quad (9)$$

(b) Mean channel (order 0). The angle-independent bracket of (7) must vanish on its own:

$$A'' + \frac{1}{r} A' + \sigma A = 0. \quad (10)$$

This is *Bessel's equation of order zero*. The solution regular at the origin is

$$A(r) = c_0 J_0(\sqrt{\sigma} r). \quad (11)$$

(c) Modulation channel (order ν). Using (8), the coefficient of Θ in (7) must vanish:

$$B'' + \frac{1}{r} B' + \left(\sigma - \frac{\nu^2}{r^2}\right) B = 0. \quad (12)$$

This is *Bessel's equation of order ν* , with regular solution

$$B(r) = c_\nu J_\nu(\sqrt{\sigma} r). \quad (13)$$

Result of the substitution. The additive ansatz solves the Helmholtz equation *if and only if* Θ is an angular eigenfunction (9) and A, B are Bessel functions of orders 0 and ν at one common σ :

$$F(r, \theta) = c_0 J_0(\sqrt{\sigma} r) + c_\nu J_\nu(\sqrt{\sigma} r) \Theta(\theta).$$

3 What this ansatz actually is

Equation (11)–(13) makes the status of the additive ansatz precise, and the conclusion is sobering: **it is not a generalization of separation of variables, it is a restriction of the full series to two**

angular harmonics. Writing the angular “mean” factor as the $\nu = 0$ eigenfunction (the constant 1) and the modulation as the ν eigenfunction, the ansatz is literally

$$F = \underbrace{c_0 J_0(\sqrt{\sigma}r) \cdot 1}_{\nu=0 \text{ product mode}} + \underbrace{c_\nu J_\nu(\sqrt{\sigma}r) \Theta(\theta)}_{\nu \text{ product mode}}, \quad (14)$$

i.e. a sum of *two* separable Helmholtz modes evaluated at the *same* eigenvalue σ . It is the two-term truncation of the Fourier–Bessel double series written in the problem statement (Eqs. (15)–(17) of `2d-problem.tex`).

The one genuinely new object relative to a single product $R(r)\Theta(\theta)$ is the **angular–mean channel** $A(r) = J_0(\sqrt{\sigma}r)$. Over the wedge,

$$\frac{N}{2\pi} \int_0^{2\pi/N} \Theta(\theta) d\theta = 0 \quad \text{when } \nu = nN,$$

so $A(r)$ is exactly the angular average of F and $B(r)\Theta(\theta)$ is the zero-mean fluctuation. This is the analytic content of the empirical decomposition: $A \leftrightarrow g(r)$ (the dominant symmetric profile) and $B\Theta \leftrightarrow a(r) \cos N\theta$ (the small modulation). The additive ansatz is therefore the *right bookkeeping* for the numerics — but, as the next section shows, it inherits the same fatal incompatibility as the product ansatz, and it makes the source of that incompatibility unusually visible.

4 Boundary conditions

4.1 Outer absorbing condition (i): the channels disagree on σ

Condition (i) requires $F(1, \theta) = 0$ for *every* $\theta \in [0, 2\pi/N]$. With (5),

$$A(1) + B(1) \Theta(\theta) = 0 \quad \text{for all } \theta. \quad (15)$$

Because $\Theta(\theta)$ is non-constant, this splits into *two* independent Dirichlet conditions,

$$A(1) = 0 \quad \text{and} \quad B(1) = 0, \quad (16)$$

that is,

$$J_0(\sqrt{\sigma}) = 0 \quad \text{and} \quad J_\nu(\sqrt{\sigma}) = 0. \quad (17)$$

Each channel quantizes σ on its own grid,

$$\text{mean channel: } \sqrt{\sigma} \in \{\kappa_{0,j}\}, \quad \text{modulation channel: } \sqrt{\sigma} \in \{\kappa_{\nu,k}\}, \quad (18)$$

where $\kappa_{m,\ell}$ is the ℓ -th positive zero of J_m . For a single shared σ with *both* $c_0 \neq 0$ and $c_\nu \neq 0$, (17) demands a *common positive zero* of J_0 and J_ν .

Bourget’s obstruction. Bessel functions of the first kind of distinct integer orders share no positive zeros — Bourget’s hypothesis, proven by Siegel. (Here $\nu = nN$ with $n \geq 1$ from periodicity, so the orders 0 and ν are distinct integers.) Moreover, for real order all zeros of J_ν are real, so allowing complex σ does not manufacture a coincidence either. Hence (17) has no solution with both channels active:

No exact additive eigenmode $A + B\Theta$ with $A, B \neq 0$ exists.

(19)

The angularly symmetric profile wants to decay at rate $\kappa_{0,j}^2$; the N -fold modulation wants to decay at rate $\kappa_{N,k}^2 \neq \kappa_{0,j}^2$. A single exponential $e^{-\sigma t}$ cannot serve both.

Why this is the useful part. In `analytical_attempt_coupled_DL_AL.tex` the failure of separation was diagnosed at the MT condition (the explicit $2/r$ making the angular eigenvalue r -dependent, and $R \not\propto \rho$). The additive ansatz relocates the *first* obstruction to the *outer* boundary: the two channels are already incompatible at $r = 1$, by Bourget, *before* the microtubule physics is consulted. The incompatibility is thus not an artifact of the MT jump — it is baked into the requirement that one decay rate govern both the mean and the modulation.

4.2 Periodicity (ii)

With $\Theta = \alpha \cos \nu \theta + \beta \sin \nu \theta$, condition (ii) $\Theta(0) = \Theta(2\pi/N)$ gives

$$\alpha \left(1 - \cos \frac{2\pi\nu}{N}\right) = \beta \sin \frac{2\pi\nu}{N}, \quad (20)$$

the value-periodicity already discussed in the audit note. The smooth choice $\nu = nN$ satisfies it; this is what makes 0 and ν distinct integer Bessel orders in (17) and locks in the Bourget obstruction.

4.3 Microtubule condition (iii): a second over-determination

Even setting aside §4.1, condition (iii) cannot be met by (5) for a single σ . Because $R[\cdot]$ in (2) is linear and $F(r, 0) = A(r) + \Theta(0)B(r)$,

$$R(r) = R[A](r) + \Theta(0) R[B](r), \quad (21)$$

and (iii) reads, for all $r \in [0, 1]$,

$$\underbrace{a A - b R[A]}_{\text{mean part}} + \Theta(0) \underbrace{(a B - b R[B])}_{\text{modulation part}} = \frac{2 \Theta'(0)}{r} B(r). \quad (22)$$

The functions $\{A, R[A], B, R[B], B/r\} = \{J_0, R[J_0], J_\nu, R[J_\nu], J_\nu/r\}$ are linearly independent on $[0, 1]$ — J_0 and J_ν solve different-order Bessel equations, and $R[\cdot]$ carries the advective exponential $e^{(b-\sigma)r/v}$ which is neither. No choice of the finite parameter set $\{c_0, c_\nu, \alpha, \beta, \sigma\}$ makes the combination (22) vanish identically in r . In particular the explicit $2/r$ on the right (the geometric coefficient inherited from the $1/r^2$ angular Laplacian and the $\delta(\theta - \theta_i)/r$ source) cannot be matched by the exponential/Bessel structure on the left at a single σ . This is the same r -dependent over-determination found for the product ansatz, now appearing as the failure to balance (22).

5 The resolution the ansatz points to

The additive ansatz is not a dead end — it is the correct *trial space*, once we stop insisting on a single real σ . The two obstructions above say only that the mean and modulation channels cannot share a real Dirichlet-zero decay rate. Replace the eigenvalue $-\sigma$ by the Laplace variable s and *couple* the channels through the boundary data instead of through a shared Bessel zero:

$$F(r, \theta; s) = A(r; s) + B(r; s) \Theta(\theta), \quad (23)$$

where A solves the order-0 and B the order- ν *modified*-Bessel equation (argument $\sqrt{s}r$), each with its own radial Green's function, and the microtubule condition (22) ties the two amplitudes together. Projecting (22) onto the lowest radial profile of each channel collapses it to a 2×2 secular block

$$\det \begin{bmatrix} a p_{00}(s) - b q_{00}(s) & a p_{0N}(s) - b q_{0N}(s) \\ a p_{N0}(s) - b q_{N0}(s) - \frac{2N}{\cdot} \ell_N(s) & a p_{NN}(s) - b q_{NN}(s) \end{bmatrix} = 0, \quad (24)$$

whose slowest root $s_* = -\sigma_*$ is the leading (generally complex) decay rate. This is *exactly* the two-mode reduction ($n \in \{0, \pm 1\}$, orders 0 and N) derived in §5 of `analytical_attempt_coupled_DL_AL.tex`. The additive ansatz is its analytic skeleton: the channel $A = J_0$ is the symmetric $g(r)$, the channel $B\Theta = J_N \cos N\theta$ is the modulation $a(r) \cos N\theta$, and the eigenvector of (24) fixes the small amplitude ratio c_ν/c_0 that the affine-overlap test measured numerically.

6 Summary

- Substituting $F = A(r) + B(r)\Theta(\theta)$ into $-\sigma F = \nabla^2 F$ *separates cleanly*: it forces $\Theta'' + \nu^2 \Theta = 0$, sends A to Bessel order 0 ($A = J_0(\sqrt{\sigma}r)$) and B to Bessel order ν ($B = J_\nu(\sqrt{\sigma}r)$), both at one common σ .

- Consequently the additive ansatz is *not* more general than separation of variables; it is the two-term Fourier–Bessel sum (the $\nu = 0$ and ν modes). Its only new ingredient is the angular–mean channel $A = J_0$, the analytic counterpart of the numerically dominant $g(r)$.
- The outer absorbing condition (i) alone forces $J_0(\sqrt{\sigma}) = 0$ *and* $J_\nu(\sqrt{\sigma}) = 0$ — a common positive zero of two distinct-order Bessel functions, which Bourget’s theorem forbids. The mean channel and the modulation channel demand incompatible decay rates, so no exact single- σ additive eigenmode exists.
- This relocates the obstruction to the *outer* boundary, ahead of the MT condition; (iii) then over-determines a second time, since $\{J_0, R[J_0], J_\nu, R[J_\nu], J_\nu/r\}$ are independent in r and cannot cancel at one σ .
- Relaxing “single real σ ” to a Laplace variable s keeps the additive structure and couples the two channels through the MT condition, yielding the 2×2 secular determinant (24) — precisely the working two-mode reduction. The additive ansatz is the right trial space for that reduction and reproduces the observed $g(r) + a(r) \cos N\theta$ form.

Honest status. As with the product ansatz, the additive ansatz admits *no* exact closed-form single- σ eigenfunction; the obstruction is structural (two channels, two incompatible Bessel-zero spectra, tied by an r -dependent boundary functional). What it *does* provide is the cleanest statement of *why* — Bourget incompatibility at $r = 1$ — and the correct two-channel basis $\{J_0, J_N\Theta\}$ for the resolvent/secular construction.